A Benchmark for Human Personality Analysis in Multimodal Dialogues

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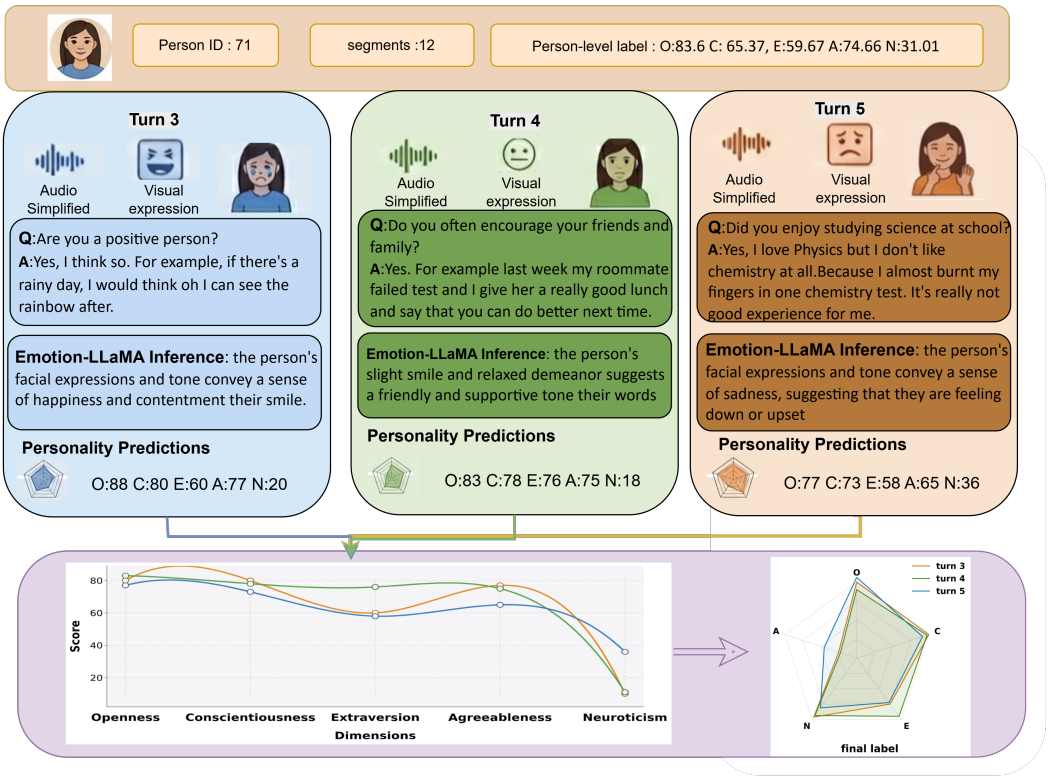


Figure 1: An example of turn-level multimodal personality analysis in an interview conversation.

Abstract

Human personality analysis is crucial for personalized dialogue systems, mental health assessment, and human–computer interaction, but existing resources are mostly text-only and fail to capture

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rich cognitive cues (e.g., speaker-awareness and emotions) in real conversations. We present a new benchmark for human personality analysis in multimodal dialogues (HPAMD), built on 218 real-world IELTS-style interview videos (RIIV) and segmented into 2,322 turn-level units with aligned emotion-related dialogue text, acoustic, visual, and continuous Big Five annotations. We further propose DAMP, a dialogue-aware multimodal framework that models turn-level question-response interactions and integrates emotion reasoning for personality prediction. Experiments show dialogue modeling outperforms whole-video modeling, emotion-enhanced representations bring additional gains, and DAMP achieves the best overall performance under the full setting. The dataset is publicly available at: https://github.com/lulvhui/RIIV.

CCS Concepts

• **Applied computing** → *Psychology*; • **Computing methodologies** → **Discourse, dialogue and pragmatics**; **Supervised learning by regression**.

Keywords

Human personality analysis, Multimodal benchmark, Multimodal dialogues, Dialogue modeling, Emotion-enhanced

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1 Introduction

Human personality shapes behavior, communication styles, and social interactions, and is widely applied in personalized dialogue systems, mental health assessment, etc. Psychological research confirms personality traits are reflected in linguistic patterns, vocal prosody, and facial expressions [15, 23], inspiring studies to infer psychological profiles from user data [13, 17, 24, 27].

However, existing human personality analysis (HPA) works are largely limited to text-based datasets [7, 32], which overlook critical multimodal behavioral signals for real-world personality perception. Though some studies explore multimodal dialogues for HPA (**HPAMD**), they focus on human-agent virtual dialogues rather than real conversations and ignore turn-level dialogue information [28]. This oversight can lead to inaccurate predictions. For instance, in Figure 1, turn 3 and turn 4 in a dialogue may convey contradictory Neuroticism scores (20 and 18) compared to turn 5 (36); mixing all turns would likely yield an unduly low Neuroticism score (below 20), while differentiated turn-level modeling can approach the correct result (about 30) by weighting the higher score from turn 5.

To address this gap, we construct a multimodal personality benchmark based on real-world IELTS-speaking interview videos (**RIIV**). Our pipeline includes three steps: collecting and cleaning high-quality interview videos with aligned transcripts, obtaining continuous Big Five annotations via human-AI collaboration, and segmenting videos into turn-level units enriched with multimodal and emotion-related information for fine-grained analysis.

Text-based and general multimodal methods are unsuitable for multimodal HPA, as they either ignore non-linguistic cues or focus on transient affective states rather than stable personality traits [3]. Existing specific multimodal HPA approaches also lack utterance-level dialogue modeling. Thus, we propose a dialogue-aware multimodal personality analysis model (**DAMP**). It constructs turn-level multimodal representations by aligning interviewers' questions and candidates' responses with multimodal signals, incorporates emotion reasoning features, and uses a dialogue-aware fusion architecture to model local and global dependencies, finally aggregating turn-level predictions for personality profiles.

Our main contributions are: (1) **Constructing a new multimodal personality benchmark** based on real IELTS-style interview videos, with aligned multimodal, turn-level, and emotion-related information plus continuous Big Five annotations. (2) **Proposing DAMP**, which explicitly models turn-level question-response interactions and integrates emotion-aware cues. (3) **Conducting extensive experiments** that verify the effectiveness of the benchmark and model, showing dialogue modeling and emotion-enhanced representations improve prediction performance.

2 Related Work

Textual and Multimodal Personality Analysis. Early personality prediction primarily relied on textual cues, including psycholinguistic features, handcrafted representations, and pretrained language models [9, 26]. Textual benchmarks such as PANDORA and Big-Five Backstage have advanced this direction, but cannot capture non-verbal and paralinguistic behaviors [20, 29].

Recent studies therefore incorporate audio and visual signals across monologues, social interactions, and structured interviews [6, 14, 16, 18, 21, 28]. However, most benchmarks remain limited to controlled settings, monologues, or virtual interactions, lacking the spontaneous turn-level dynamics of authentic dialogues.

Unlike prior works, we focus on real human personality analysis in real-world multimodal dialogues (HPAMD), and establish the benchmark with a new dedicated dataset and a tailored model.

Multimodal Dialogue Analysis. Multimodal dialogue analysis has been extensively studied for dialogue tasks including emotion recognition, sentiment analysis, and emotion cause analysis [5]. Such approaches typically fuse textual, acoustic, and visual features to model speaker states and contextual dependencies, as seen in *MKE-IGN* (advancing recognition via knowledge-enhanced interactive graph modeling [25, 30]), and the *SemEval-2024 Task 3* benchmark targeting multimodal emotion-cause extraction in dialogues [31].

However, these methods are predominantly built for short-term affective understanding, with prediction targets limited to utterance-level emotion labels or emotion-cause pairs [4, 33], focusing on transient emotional states rather than stable personality traits. In contrast, our personality analysis task requires reasoning over long-form interview videos, where personality unfolds across multiple dialogue turns and is co-shaped by candidates and examiners. Existing methods cannot directly model the required long-range behavioral consistency, turn-level interaction structure, and continuous trait regression, motivating a dialogue-aware multimodal framework tailored explicitly for personality analysis.

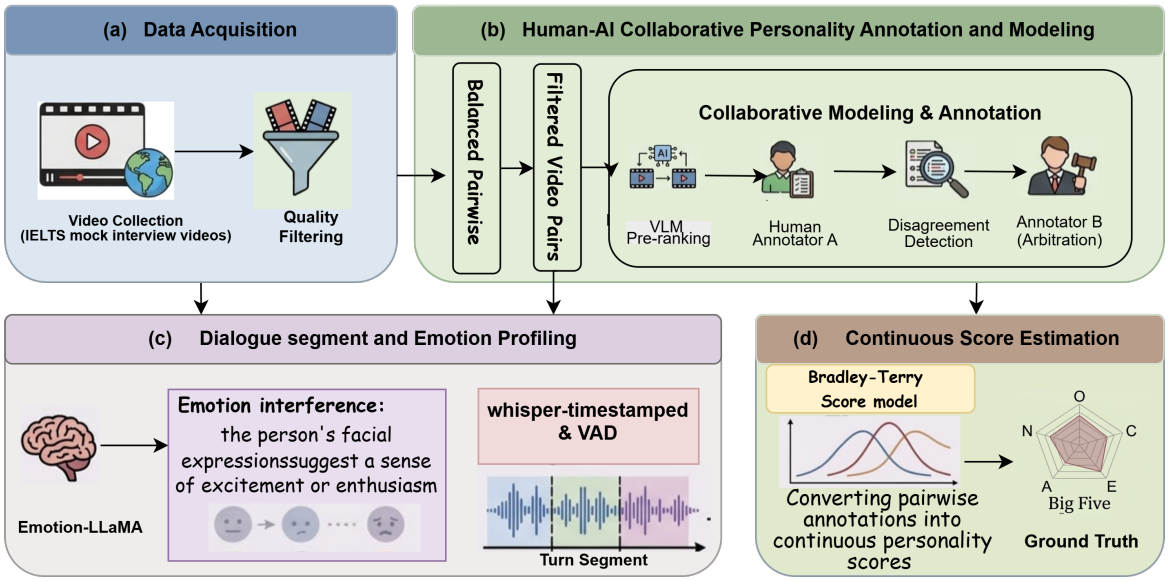


Figure 2: The overview of the construction pipeline for the proposed multimodal personality dataset RIIV.

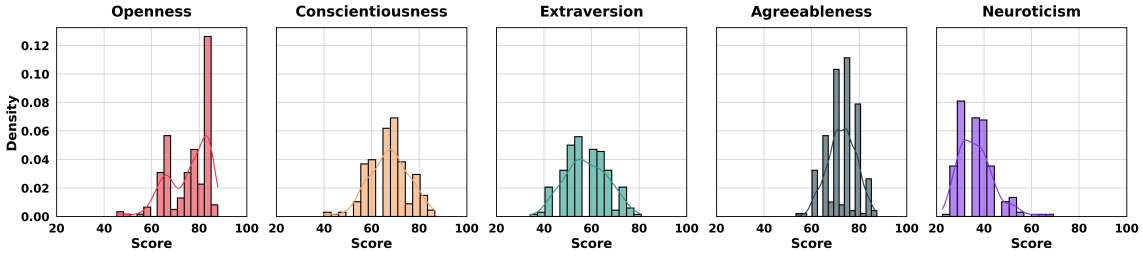


Figure 3: Continuous score distributions of the Big Five personality traits.

3 Dataset Construction

We construct a multimodal personality benchmark from real-world IELTS-speaking interview videos (**RIIV**). The pipeline consists of three stages: data collection and cleaning, human-AI collaborative annotation and turn-level dialogue segmentation with aligned emotion profiling. Figure 2 illustrates the overall workflow.

3.1 Data Collection

To capture authentic conversational behavior, we employ real-world mock IELTS-speaking videos from the psychological research project, which are collected by Southern Jiangsu Psychological Research Lab, China. We focus on Part 1 of the IELTS speaking test, where candidates answer short daily-life questions. Compared with later sections, this part involves less preparation and cognitive load, making it more likely to reveal spontaneous facial expressions, natural prosody, and interactional behavior relevant to personality analysis. After filtering out low-quality samples with severe noise, occlusion, or poor visual quality, the final dataset contains **218** high-quality interview videos.

3.2 Human-AI Collaborative Annotation

Apparent personality assessment [21] is inherently subjective, making consistent absolute scoring difficult. We therefore adopt a human-AI collaborative annotation pipeline based on pairwise comparison to derive apparent Big Five scores for the 218 videos.

Video pairs are constructed according to transcript similarity, prioritizing behaviorally contrasting samples while maintaining balanced coverage. The resulting graph contains **545** unique pairs covering all **218** subjects, with each subject involved in five comparisons. It is balanced and connected, with no self-comparisons or duplicate pairs. Each pair is evaluated on all five traits, yielding **2,725** trait-level judgments.

As shown in Figure 2(b), Qwen3-VL-Plus first generates comparative judgments from transcripts and sampled video frames, which are then verified by a human annotator through a forced-choice interface. Disagreements between the AI and the first annotator occurred in 186 pairs, covering 300 trait-level comparison instances. These disputed cases were further reviewed by a second annotator, whose decisions agreed with the first annotator in 267 cases (89.0%)

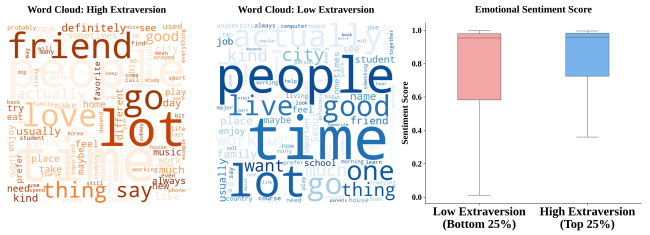


Figure 4: Textual modality analysis. Left: frequent vocabulary usage. Right: sentiment score distributions across trait groups.

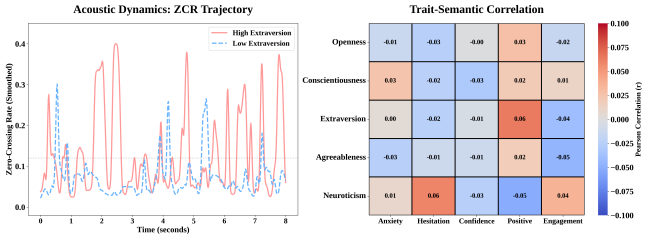


Figure 5: Acoustic modality analysis. Left: smoothed ZCR trajectories. Right: correlations between traits and acoustic-semantic cues.

and with the AI in 33 cases (11.0%). Disagreements were most frequent on openness, neuroticism, and conscientiousness, suggesting that these traits are relatively harder to judge consistently.

The verified comparisons are converted into continuous personality scores using the Bradley–Terry model [2] and normalized to $[0, 100]$ for downstream regression.

3.3 Dialogue Segmentation and Emotion Profiling

Global video-level samples are insufficient to capture fine-grained behavioral dynamics of personality expression. To support fine-grained multimodal modeling, each interview video is further segmented into turn-level dialogue units.

We first apply *whisper-timestamped* [11, 22] to obtain token-level timestamps, and then separate interviewer questions from candidate responses using VAD and speaker embeddings. The resulting boundaries are manually reviewed and corrected, producing **2,322** clean dialogue turns. For each turn, Emotion-LLaMA [3] generates a structured affective description from the aligned audiovisual clip and transcript, covering facial, vocal, and linguistic cues, as illustrated in Figure 2(c).

3.4 Dataset Statistics and Comparison

The proposed benchmark contains **218** interview videos with continuous apparent Big Five annotations, segmented into **2,322** turn-level multimodal dialogue units. Each turn is aligned with emotion-related dialogue text, acoustic, and visual information. We adopt a strict subject-level split with no subject overlap or unmatched turns: the training set contains 174 subjects and 1,832 turns, the validation set contains 22 subjects and 227 turns, and the test set contains 22 subjects and 263 turns.

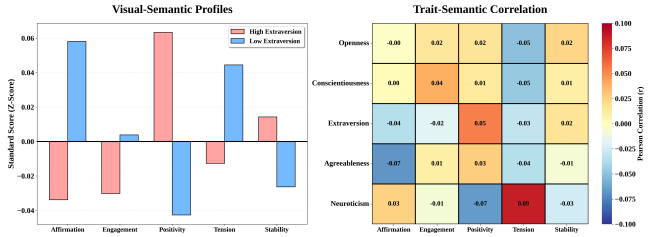


Figure 6: Visual modality analysis. Left: standardized visual-semantic profiles. Right: trait–semantic correlation heatmap.

Dataset Statistics and Modality Analysis. Figure 3 shows the distribution of the continuous personality scores derived from the Bradley–Terry model. All five traits exhibit approximately bell-shaped distributions without severe skewness, indicating balanced coverage of personality intensities.

Figures 4, 5, and 6 further reveal distinguishable personality-related patterns across textual, acoustic, and visual modalities, indicating their complementary value for personality analysis. In particular, higher extraversion is generally associated with more positive and socially engaging language, stronger prosodic variation, and more expressive emotional cues. These results suggest that multimodal signals provide complementary evidence for personality analysis.

Comparison with Existing Personality Datasets. Table 1 compares our benchmark with representative text-based and multimodal personality datasets in terms of interaction setting, modality coverage, and annotation protocol. Existing text-based resources mainly focus on written essays or social media content, while existing multimodal datasets are often limited to monologue scenarios, controlled laboratory settings, or synthetic conversations. In contrast, our dataset captures authentic dyadic interview interactions and provides aligned emotion-related dialogue text, acoustic and visual together with continuous apparent personality annotations, making it more suitable for fine-grained multimodal personality analysis.

4 Methodology

To model personality traits from conversational interactions, we propose **DAMP**, a dialogue-aware multimodal framework that performs hierarchical cross-modal fusion over aligned question–response turns and aggregates turn-level predictions into subject-level personality scores, as illustrated in Figure 7.

Task Definition. We formulate multimodal personality analysis as a person-level trait estimation problem from multi-turn interviewer-interviewee interactions. For the t-th turn of subject p, we denote the interviewer question and candidate response as $(q_t^{(p)}, s_t^{(p)})$, with aligned audio and visual observations $\mathbf{u}_t^{(p,\text{aud})}$ and $\mathbf{u}_t^{(p,\text{vis})}$, together with an auxiliary affective description $r_t^{(p)}$. The multimodal input is defined as

$$\mathcal{X}_t^{(p)} = \left(q_t^{(p)},\, s_t^{(p)},\, \mathbf{u}_t^{(p,\text{aud})},\, \mathbf{u}_t^{(p,\text{vis})},\, r_t^{(p)}\right). \tag{1}$$

Table 1: Comparison of our dataset with representative text-based and multimodal personality benchmarks. (A: Audio, V: Visual, T: Textual, Bio: Physiological, Emo: Emotion-related information).

Dataset	Scenario / Source	Label Form	Size	Modality
Essays Dataset [19]	Essays (Monologue)	Self-report	2,468 essays	T
Pandora Dataset [32]	Reddit Posts (Monologue)	Self-report	10,000 users	T
Bilingual Dialogue [10]	Synthetic (Dyadic Dialogue)	Preset Personality	12,361 utterances	T
First Impressions V2 [21]	YouTube Vlogs (Monologue)	Human Annotation	10,000 clips	A, V, T
AMIGOS [16]	Affect Elicitation	Self-report	40 participants	A, V, Bio
VPTD [6]	Video Interviews (Dyadic)	Self-report	38 interviews	Face (V)
NTT Dataset [14]	Self-introduction Videos (Monologue)	Human Annotation	10,100 videos	A, V, T
ECNU Dataset [28]	Virtual Human-Agent Interviews	Self-report	287 videos	A, V (incl. Pose), T
Ours	**Real Oral Tests (Student-Teacher Turn-level)**	**Human-AI Collaboration**	**218 videos / 2,322 turns**	**A, V, T, Emo**

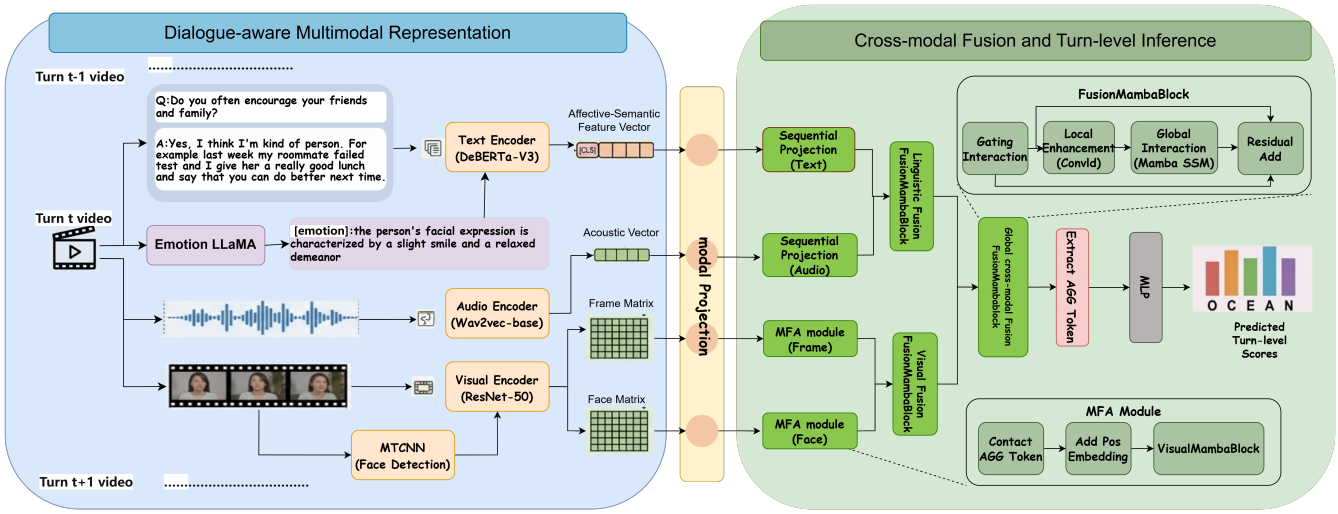


Figure 7: The overall architecture of the proposed dialogue-aware multimodal framework.

Given $\mathcal{X}_t^{(p)}$, the model predicts a turn-level trait vector

$$\hat{\mathbf{y}}_t^{(p)} = \mathcal{F}(\mathcal{X}_t^{(p)}; \Theta), \tag{2}$$

where the five dimensions correspond to Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Since annotations are defined at the subject level, the final prediction for subject p is obtained by averaging the predictions over all N_p turns:

$$\bar{\mathbf{y}}^{(p)} = \frac{1}{N_p} \sum_{t=1}^{N_p} \hat{\mathbf{y}}_t^{(p)}. \tag{3}$$

Dialogue-aware Multimodal Modeling.

For **each conversational turn**, we employ pre-trained unimodal encoders to obtain four representations: (1) a text representation derived from the interviewer question, candidate response, and **emotion-reasoning** text (as emotion feature injection); (2) an audio representation capturing prosodic and paralinguistic cues from speech; (3) a frame-level visual representation encoding global behavioral information; and (4) a face-level visual representation modeling localized facial expressions.

These features are projected into a shared latent space and fused hierarchically. Text and audio are first combined, while frame- and face-level visual features are integrated separately. The resulting representations are then fused by a *FusionMamba* block following Tang et al. [28], whose gated state-space modeling captures complementary cross-modal and temporal dependencies while suppressing redundancy.

The fused turn representation $\mathbf{z}_t$ is passed to a prediction head to produce personality scores: $\hat{\mathbf{y}}_t = \mathcal{G}(\mathbf{z}_t)$. During training, each turn is supervised by the subject-level annotation; during inference, turn-level predictions are averaged to obtain the final subject-level personality profile.

5 Experiments

To verify the HPAMD benchmark we established, we conduct experiments using recent and best-performing models for HPA.

5.1 Baselines and Metrics

We compare the proposed **DAMP** with one multimodal large model baseline and three representative multimodal affective and personality analysis methods. 1) **Qwen3-VL-8B-Instruct** [1], the most representative and competitive multimodal large model. 2) **KANMCP**

Table 2: Performance comparison of different methods for HPA. The best results are in bold.

Method	Setting	O	C	E	A	N	Avg
Qwen3-VL-8B [1]	Zero-shot	0.9217	0.9088	0.9031	0.9279	0.9216	0.9166
	Few-shot	0.9277	0.8977	0.9296	0.9566	0.8976	0.9218
	SFT	0.9190	0.9338	0.9172	0.9545	0.9290	0.9307
KANMCP [12]	SFT	0.8852	0.9146	0.9126	0.9272	0.9339	0.9147
TRD [8]	SFT	0.9303	0.9330	0.9295	0.9516	0.9454	0.9379
PINet [28]	SFT	0.9241	0.9280	0.9385	0.9520	**0.9528**	0.9391
DAMP (Ours)	SFT	**0.9597**	**0.9384**	**0.9563**	**0.9576**	0.9475	**0.9519**
w/o Emotion	SFT	0.9526	0.9382	0.9459	0.9566	0.9477	0.9482

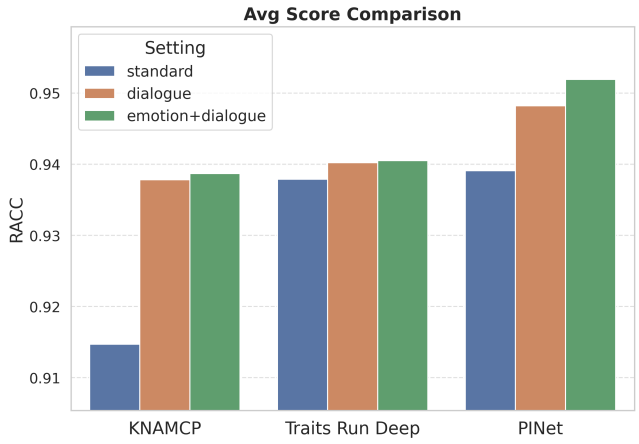


Figure 8: Previous approaches for HPA (KANMCP, Traits Run Deep, and PINet) under our designed dialogue modeling and emotion injection.

[12], the SOTA for multimodal emotion detection with utterance-level modeling. 3) **Traits Run Deep (TRD)** [8], a personality-oriented multimodal framework. 4) **PINet** [28], the SOTA for HPA in dialogues, but largely resorts to human pose features.

We adopt **Mean Accuracy** as the evaluation metric, defined as $Acc = 1 - \text{MAE}$, where MAE is the mean absolute error between predicted and apparent Big Five scores within RIIV. We report the performance on the five Big Five dimensions: **Openness (O)**, **Conscientiousness (C)**, **Extraversion (E)**, **Agreeableness (A)**, and **Neuroticism (N)**, together with their average score (**Avg**).

5.2 Results and Analysis

Comparison with SOTAs. Table 2 compares different methods for multimodal HPA. Despite its large scale, Qwen3 remains limited on this task, while KANMCP achieves competitive performance through utterance-level modeling, highlighting the importance of turn-level dialogue information. TRD and PINet perform better overall but do not jointly model dialogue structure and emotion cues. By addressing both aspects, DAMP achieves the best overall performance.

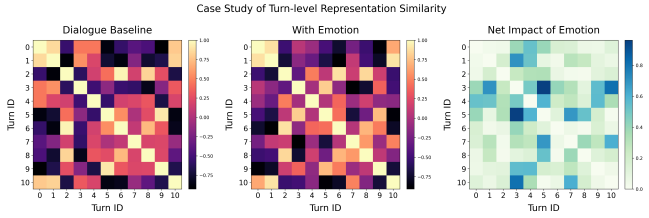


Figure 9: Case study of turn-level representation similarity

Impact of dialogue modeling and emotional injection for HPAMD. As shown in Figure 8, all existing methods achieve consistent performance improvements when integrated with our proposed dialogue modeling module, verifying the effectiveness of our dialogue modeling strategy. Notably, our emotional information is aligned with each dialogue turn, meaning emotional cues can only be effectively incorporated on the basis of dialogue modeling. Accordingly, applying our designed emotional injection mechanism to existing methods also yields consistent performance gains, further demonstrating the validity of this mechanism.

Visualization of our benchmark value with emotion cues. As shown in Figure 9, we visualize turn-level representation similarity to examine the impact of emotion features. The left Dialogue Baseline panel shows a dispersed similarity structure without clear personality-aligned clustering. In contrast, the middle With Emotion panel exhibits a more compact, diagonal-concentrated structure with clearer separation between similar and dissimilar turns, suggesting that emotion reasoning better captures consistent personality traits. The right Net Impact of Emotion panel further shows suppressed spurious similarities across different personality profiles and reinforced coherence for turns from the same individual. Overall, emotion features yield more personality-discriminative turn-level representations.

6 Conclusion

In this work, we introduce a new multimodal benchmark for human personality analysis in multimodal dialogues (HPAMD). Based on this benchmark, we develop a dialogue-aware framework (DAMP) that models turn-level interactions and incorporates emotional reasoning. Experimental results demonstrate the value of the benchmark and show that both dialogue structure and affective cues are important for accurate personality prediction.

Acknowledgments

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